Clinical & Causal Evaluation Report

Study: CAUSAL-EHR-2026-VASO — effect of early_vasopressor on mortality_28d (risk_difference)

Population: adult septic shock, first ICU stay

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Data mode: synthetic validation — oracle-anchored, estimators blinded to ground truth.

1. Executive Summary & Decision Analysis

- Pipeline Decision: NO-GO
- Decision driver: Safety guardrail, not efficacy. The efficacy endpoint passed (DML ATE -5.03pp, 95% CI [-6.70pp, -3.35pp], meets the pre-registered ≤ -3.00pp threshold). The verdict is overturned by guardrail breach(es): acute_kidney_injury (+3.7pp).

Key Findings — Oracle vs. Estimators. Ground-truth effect is a -5.86pp absolute change (NNT ≈ 17). Both causal estimators recover the correct sign and clinically concordant magnitude (DML -5.03pp; IPW -4.02pp).

The naive association points the opposite way (+2.55pp) — it would misclassify the intervention's direction of effect.

> CRITICAL — Sign discordance. The unadjusted signal and the causal truth disagree in direction, not merely magnitude. Any decision layer consuming observational contrasts would reach the clinically inverted conclusion.

Clinical Rationale. The intervention is causally efficacious and survives refutation (E-value 1.66); deployment is withheld because a pre-committed safety guardrail is breached. Correct posture: conditional pause — pursue heterogeneity analysis to find the sub-population in whom benefit dominates the safety cost.

2. Confounding & Discrepancy Diagnostics

- Bias Magnitude: |Naive − Oracle| = |(+2.55pp) − (-5.86pp)| = 8.41pp (≈ 1.43× the reference effect size).
- Simpson's Paradox / severe selection bias Detected: YES (sign reversal between naive and oracle).
- Treatment predictability (propensity AUC): 0.74 — strong confounding by baseline covariates.

Clinical Mechanism — confounding by indication. Clinicians preferentially initiate treatment in the sickest patients (elevated severity, higher baseline risk). Treatment is therefore correlated with the outcome through the shared common cause of illness severity, independent of any drug effect. In the unadjusted contrast this baseline-risk imbalance dominates and inverts the true effect, manufacturing a false signal. Conditioning on the backdoor set — verified by post-weighting balance (all |SMD| < 0.10) — removes the spurious component.

> WARNING — adjustment must be orthogonalized. A high propensity AUC confirms treatment is strongly predictable from severity; naïve regression that is not cross-fitted/orthogonalized remains exposed to residual regularization bias in this regime.

3. Causal Estimator Performance Benchmarking

Estimator	Estimated ATE	Error vs. Reference	Relative Performance
Oracle (Ground Truth)	-5.86pp	0.0000	Benchmark
DML	-5.03pp	0.84pp (14.3% rel.)	Rank 1 — best. Closest recovery, CI excludes null
IPW	-4.02pp	1.85pp (31.5% rel.)	Rank 2. ~2.2× the DML error
Naive Association	+2.55pp	8.41pp (bias)	Unadjusted — directionally wrong

Methodological Takeaway — why DML leads. Double Machine Learning estimates the effect from residual-on-residual regression after flexibly modeling both E[Y|W] and E[T|W]; the Neyman-orthogonal moment condition is first-order insensitive to nuisance error and cross-fitting removes overfitting bias. It is doubly robust — consistent if either nuisance model is approximately correct — whereas IPW depends solely on a correct propensity model and suffers variance inflation from extreme inverse-propensity weights near the overlap boundary (mitigated but not eliminated by stabilization and 2–98% trimming). IPW is retained as a transparent, assumption-light cross-check: sign/magnitude agreement between the two is itself evidence no single functional form drives the result.

4. Next Steps & Actionable Recommendations

1. Subgroup analysis (CATE). Estimate individualized effects with CausalForestDML / T-/X-learner on the pre-declared effect modifiers (sofa_score, lactate). Objective: convert a population-level NO-GO into a targeted, guardrail-compliant policy by locating the strata where benefit exceeds the safety cost; evaluate via a policy-value / uplift curve.

2. Sensitivity testing. Report the E-value (1.66 point / 1.48 at the CI bound): the confounder–treatment and confounder–outcome association strength required to nullify the effect. Complement with Rosenbaum Γ-bounds and the add_unobserved_common_cause simulation sweep.

3. Model validation. Confirm propensity overlap / common support (histograms, 0.05–0.95 trimming), re-check covariate balance (|SMD| < 0.10) after weighting, and add outcome-model calibration (Brier, calibration slope) plus a negative-control outcome. Strata outside common support are non-identifiable and must be excluded.

> BOTTOM LINE. Causally effective (NNT ≈ 17) and robust, but deployment is halted on the safety guardrail. Immediate path forward is CATE-driven patient targeting, not efficacy re-litigation.

Appendix — Clinical risk of trusting observational metrics

Relying on the naive association (+2.55pp) instead of the causal estimate would have judged the intervention harmful and withheld a therapy that changes the outcome by -5.86pp absolute (≈ 6 events per 100 patients; NNT ≈ 17). The observational metric recommends the opposite clinical action

— the defining hazard of correlational analytics under confounded treatment assignment.